Indices, Surds and Polynomials

Section 1: Fractional Indices and Exponential Equations

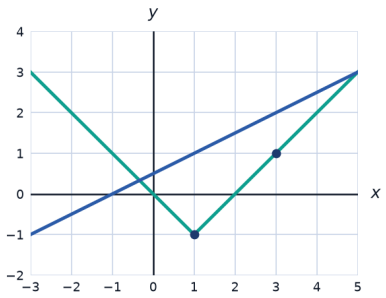
1 Evaluate the exact numerical value of $16^{-\frac{3}{4}} \times 27^{\frac{2}{3}}$.

a. $\frac{9}{8}$

b. $\frac{8}{9}$

c. $\frac{3}{4}$

d. $\frac{27}{16}$



2 Solve the exponential equation $2^{2x+1} - 9(2^x) + 4 = 0$ for all real x.

a. $x = -1$ and $x = 2$

b. $x = 1$ and $x = 4$

c. $x = 0$ and $x = 2$

d. $x = -2$ and $x = 1$

3 Simplify the index expression $\frac{(4x^3y^{-2})^{\frac{1}{2}}}{2x^{-1}y^3}$.

a. $x^{\frac{5}{2}}y^{-4}$

b. $2x^{\frac{5}{2}}y^{-4}$

c. x^2y^{-4}

d. $x^{\frac{5}{2}}y^2$

Section 2: Surds and Rationalising Denominators

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4 Rationalise the denominator and simplify $\frac{4+\sqrt{3}}{2-\sqrt{3}}$ into the form $a+b\sqrt{3}$.

a. $11+6\sqrt{3}$

b. $5+6\sqrt{3}$

c. $11-6\sqrt{3}$

d. $7+4\sqrt{3}$

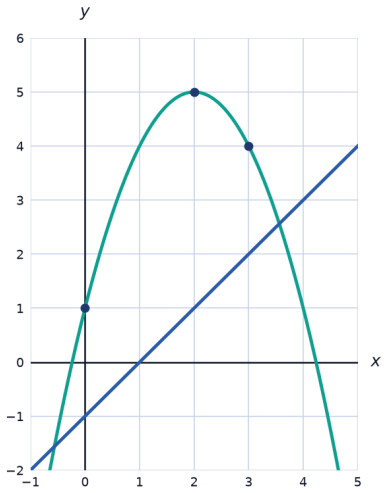
5 Express $(3-2\sqrt{5})^2$ in the simplified form $p+q\sqrt{5}$.

a. $29-12\sqrt{5}$

b. $29+12\sqrt{5}$

c. $11-12\sqrt{5}$

d. $19-6\sqrt{5}$



6 Solve the radical equation $\sqrt{2x+7}-x=2$ for all valid real solutions.

a. $x=1$ only

b. $x=1$ and $x=-3$

c. $x=3$ only

d. No real solutions

Section 3: Remainder and Factor Theorems

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7 When $P(x) = 2x^3 + ax^2 - 5x + 6$ is divided by $(x - 2)$, the remainder is 16. Find the value of a.

a. $a = 1$

b. $a = -2$

c. $a = 3$

d. $a = 4$

8 Given that $(2x - 1)$ is a factor of $f(x) = 2x^3 + 3x^2 - 8x + c$, find the constant c.

a. $c = 3$

b. $c = -3$

c. $c = 6$

d. $c = -2$

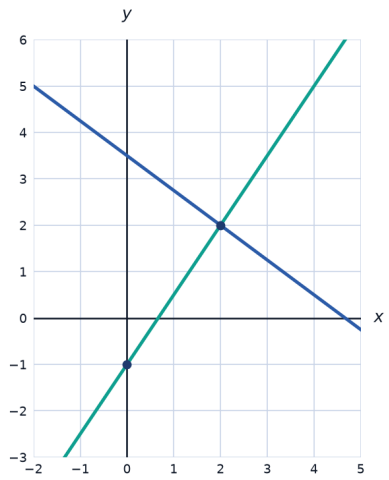
9 Factorise completely the cubic polynomial $g(x) = x^3 - 4x^2 - 7x + 10$.

a. $(x - 1)(x + 2)(x - 5)$

b. $(x + 1)(x - 2)(x + 5)$

c. $(x - 1)(x - 2)(x - 5)$

d. $(x + 1)(x + 2)(x - 5)$



Section 4: The Binomial Theorem

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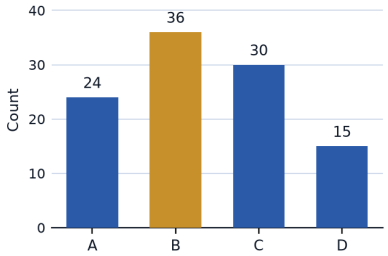
10 Find the coefficient of x^3 in the binomial expansion of $(1+2x)^6$.

a. 160

b. 80

c. 240

d. 120



11 Find the term independent of x (the constant term) in the expansion of $\left(x^2+\frac{2}{x}\right)^6$.

a. 240

b. 160

c. 60

d. 192

12 In the expansion of $(1+ax)^5(1-x)$, the coefficient of x is 9. Find the value of a.

a. $a=2$

b. $a=1$

c. $a=3$

d. $a=-2$

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